

## Advanced Proof Patterns

### Example 1 : Proof by Cases ( Three - Way Split )

Prove: For all integers  $n$ ,  $n \langle 0(or)(n) = 0(or)(n) \rangle 0$

$$\frac{}{n < 0 \vee n = 0 \vee n > 0} \text{ [trichotomy of integers, axiom]}$$

This uses the trichotomy property as an axiom. To prove something about all integers, we can case-analyze on these three possibilities.

### Example 2 : Multi - Level Case Analysis

Prove:  $(p \wedge q) \wedge (r \vee s) \Rightarrow (p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)$

$$\frac{\frac{\frac{\frac{}{p \wedge r} [\wedge\text{-intro}]}{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)} [\vee\text{-intro}]}{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)} [\vee\text{-intro}]}{\frac{\frac{\frac{\frac{}{q \wedge s} [\wedge\text{-intro}]}{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)} [\vee\text{-intro}]}{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)} [\vee\text{-intro}]}{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)} [\vee\text{-intro}]}$$

$$\frac{\frac{\frac{\frac{}{p \wedge s} [\wedge\text{-intro}]}{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)} [\vee\text{-intro}]}{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)} [\vee\text{-intro}]}{\frac{\frac{\frac{\frac{}{q \wedge r} [\wedge\text{-intro}]}{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)} [\vee\text{-intro}]}{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)} [\vee\text{-intro}]}{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)} [\vee\text{-intro}]}$$

$$\frac{\frac{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)}{((p \vee q) \wedge (r \vee s)) \Rightarrow (p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)} [\Rightarrow\text{-intro}^{[1]}]}{\frac{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)}{((p \vee q) \wedge (r \vee s)) \Rightarrow (p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)} [\Rightarrow\text{-intro}^{[1]}]}$$

Nested case analysis on two disjunctions, exploring all four combinations.

### Example 3 : Proof by Mathematical Induction ( Base land Step )

Prove: For all  $n \in \mathbb{N}$ ,  $\text{sum}(1 \text{ to } n) = n(n+1)/2$

Base case ( $n = 0$ ):

$$\frac{\frac{\frac{\frac{}{\text{true}} [\top]}{\text{sum\_to}(0) = 0} [\text{definition}]}{0 * (0 + 1) \text{ div } 2 = 0} [\text{arithmetic}]}{\frac{\text{sum\_to}(0) = 0 * (0 + 1) \text{ div } 2} {\text{true} \Rightarrow (\text{sum\_to}(0) = 0 * (0 + 1) \text{ div } 2)} [\Rightarrow\text{-intro}^{[1]}]}$$

Inductive step (assume for  $n$ , prove for  $n+1$ ):

$$\frac{\begin{array}{c} \frac{\text{sum\_to}(n) = n * (n + 1) \text{ div } 2^{\lceil 1 \rceil}}{\text{sum\_to}(n + 1) = \text{sum\_to}(n) + (n + 1)} \quad [\text{definition}] \\ \frac{\text{sum\_to}(n + 1) = n * (n + 1) \text{ div } 2 + (n + 1)}{\text{sum\_to}(n + 1) = (n * (n + 1) + 2 * (n + 1)) \text{ div } 2} \quad [\text{substitution}] \\ \frac{\text{sum\_to}(n + 1) = (n * (n + 1) + 2 * (n + 1)) \text{ div } 2}{\text{sum\_to}(n + 1) = (n + 1) * (n + 2) \text{ div } 2} \quad [\text{algebra}] \\ \text{sum\_to}(n + 1) = (n + 1) * (n + 2) \text{ div } 2 \quad [\text{factoring}] \end{array}}{(\text{sum\_to}(n) = n * (n + 1) \text{ div } 2) \Rightarrow (\text{sum\_to}(n + 1) = (n + 1) * (n + 2) \text{ div } 2)} \quad [\Rightarrow \text{-intro}^{\lceil 1 \rceil}]$$

By induction, the formula holds for all natural numbers.

### Example 4 : Structural Induction on Lists

Prove: For all sequences  $s$ ,  $\text{reverse}(\text{reverse}(s)) = s$

Base case (empty sequence):

$$\frac{\frac{\frac{\text{true}^{\neg[1]}}{\text{reverse}(\text{emptyseq}) = \text{emptyseq}} \text{ [definition]}}{\text{reverse}(\text{reverse}(\text{emptyseq})) = \text{reverse}(\text{emptyseq})} \text{ [substitution]}}{\text{reverse}(\text{reverse}(\text{emptyseq})) = \text{emptyseq}} \text{ [definition]}}{\text{true} \Rightarrow (\text{reverse}(\text{reverse}(\text{emptyseq})) = \text{emptyseq})} \text{ } [\Rightarrow \text{-intro}^{[1]}]$$

Inductive step (assume for  $s$ , prove for  $\text{cons}(x, s)$ ):

$$\frac{\frac{\frac{\text{reverse}(\text{reverse}(s)) = s \text{ [1]}}{\text{reverse}(\text{cons}(x, s)) = \text{append}(\text{reverse}(s), x)} \text{ [definition]}}{\text{reverse}(\text{reverse}(\text{cons}(x, s))) = \text{reverse}(\text{append}(\text{reverse}(s), x))} \text{ [substitution]}$$

$$\frac{\text{reverse}(\text{append}(\text{reverse}(s), x)) = \text{cons}(x, \text{reverse}(\text{reverse}(s))) \text{ [definition]}}{\text{reverse}(\text{append}(\text{reverse}(s), x)) = \text{cons}(x, \text{reverse}(\text{reverse}(s)))} \text{ [inductive hypothesis]}$$

$$\frac{\text{cons}(x, \text{reverse}(\text{reverse}(s))) = \text{cons}(x, s)}{(\text{reverse}(\text{reverse}(s)) = s) \Rightarrow (\text{reverse}(\text{reverse}(\text{cons}(x, s))) = \text{cons}(x, s))} \text{ [} \Rightarrow \text{-intro}^{[1]} \text{]}$$

By structural induction,  $\text{reverse}(\text{reverse}(s)) = s$  for all sequences.

### Example 5 : Constructive Existence Proof

Prove: There exists an even number greater than 10

$$\frac{\begin{array}{c} \frac{\text{true}}{12 = 2 * 6} \quad [\text{arithmetic}] \\ \frac{}{\text{even}(12)} \quad [\text{definition of even, } 12 = 2 * 6] \\ \frac{}{12 > 10} \quad [\text{arithmetic}] \end{array}}{\text{even}(12) \wedge 12 > 10} \quad [\wedge\text{-intro}]$$
$$\frac{\exists n : \mathbb{N} \bullet \text{even}(n) \wedge n > 10}{\text{true} \Rightarrow (\exists n : \mathbb{N} \bullet \text{even}(n) \wedge n > 10)} \quad [\exists\text{ intro with } n = 12]$$
$$\text{true} \Rightarrow (\exists n : \mathbb{N} \bullet \text{even}(n) \wedge n > 10) \quad [\Rightarrow\text{-intro}^{[1]}]$$

Constructive proof: we exhibit a specific witness (12).

### Example 6 : Non - Constructive Existence Proof

Prove: There exist irrational numbers  $a$  and  $b$  such that  $a^b$  is rational

$$\begin{array}{c}
\frac{\frac{\frac{\text{power}(\text{power}(\text{sqrt}(2), \text{sqrt}(2)), \text{sqrt}(2)) = \text{power}(\text{sqrt}(2), \text{sqrt}(2) * \text{sqrt}(2))}{\text{power}(\text{sqrt}(2), \text{sqrt}(2) * \text{sqrt}(2)) = \text{power}(\text{sqrt}(2), 2)} \quad \begin{array}{l} \text{[exponent law]} \\ \text{[arithmetic]} \end{array}}{\text{power}(\text{sqrt}(2), 2) = 2} \quad \text{[simplification]}}{\frac{\text{rational}(2)}{\text{rational}(2)}} \quad \text{[known]} \\
\frac{\text{irrational}(\text{sqrt}(2)) \wedge \text{irrational}(\text{sqrt}(2)) \wedge \text{rational}(\text{power}(\text{sqrt}(2), \text{sqrt}(2))) \quad \begin{array}{l} \text{[}\wedge\text{-intro]} \\ \text{[}\exists\text{ intro with a = b = sqrt(2)]} \end{array}}{\exists a, b \bullet \text{irrational}(a) \wedge \text{irrational}(b) \wedge \text{rational}(\text{power}(a, b))} \quad \frac{\text{irrational}(\text{power}(\text{sqrt}(2), \text{sqrt}(2))) \wedge \text{irrational}(\text{sqrt}(2)) \wedge \text{rational}(\text{power}(\text{power}(\text{sqrt}(2), \text{sqrt}(2)), \text{sqrt}(2)), \text{sqrt}(2))) \quad \begin{array}{l} \text{[}\wedge\text{-intro]} \\ \text{[}\exists\text{-intro]} \end{array}}{\exists a, b \bullet \text{irrational}(a) \wedge \text{irrational}(b) \wedge \text{rational}(\text{power}(a, b))} \quad \text{[}\vee\text{-elim]} \\
\frac{\exists a, b \bullet \text{irrational}(a) \wedge \text{irrational}(b) \wedge \text{rational}(\text{power}(a, b))}{\text{irrational}(\text{sqrt}(2)) \Rightarrow (\exists a, b \bullet \text{irrational}(a) \wedge \text{irrational}(b) \wedge \text{rational}(\text{power}(a, b)))} \quad \text{[}\Rightarrow\text{-intro}^{[1]}
\end{array}$$

Non-constructive: we don't know which case is true, but both lead to the conclusion.

### Example 7 : Proof by Strong Induction

Prove: Every natural number  $n \geq 2$  has a prime factorization

Base case ( $n = 2$ ):

$$\frac{\frac{\lceil true \rceil^{[1]}}{prime(2)} \text{ [definition]}}{prime\_factorization(2)} \text{ [trivial, singleton factorization]} \\ \frac{}{true \Rightarrow prime\_factorization(2)} \text{ [\Rightarrow -intro}^{[1]}]$$

Inductive step (assume for all  $k < n$ , prove for  $n$ ):

$$\frac{\begin{array}{c} \overline{\exists a, b \bullet 2 \leq a \wedge a < n \wedge 2 \leq b \wedge b < n \wedge n = a * b} \quad [\text{definition of composite}] \\ \overline{\phantom{\exists a, b \bullet 2 \leq a \wedge a < n \wedge 2 \leq b \wedge b < n \wedge n = a * b}} \quad [\text{strong IH, } a < n] \\ \frac{\phantom{\exists a, b \bullet 2 \leq a \wedge a < n \wedge 2 \leq b \wedge b < n \wedge n = a * b}}{\text{prime\_factorization}(a)} \quad [\text{strong IH, } b < n] \\ \frac{\phantom{\exists a, b \bullet 2 \leq a \wedge a < n \wedge 2 \leq b \wedge b < n \wedge n = a * b}}{\text{prime\_factorization}(b)} \quad [\text{multiplication of factorizations}] \\ \frac{\phantom{\exists a, b \bullet 2 \leq a \wedge a < n \wedge 2 \leq b \wedge b < n \wedge n = a * b}}{\text{prime\_factorization}(a * b)} \quad [n = a * b] \\ \frac{\phantom{\exists a, b \bullet 2 \leq a \wedge a < n \wedge 2 \leq b \wedge b < n \wedge n = a * b}}{\text{prime\_factorization}(n)} \quad [\vee\text{-elim}] \end{array}}{(n \geq 2) \Rightarrow \text{prime\_factorization}(n)} \quad [\Rightarrow\text{-intro}^{[1]}]$$

Strong induction: we assume the property for all smaller values, not just  $n-1$ .

### Example 8 : Proof Using Lemmas

Lemma 1: If  $n$  is even, then  $n^2$  is even

$$\frac{\frac{\frac{\frac{\frac{\frac{\text{even}(n)}{\exists k \bullet n = 2 * k} [\text{definition of even}]}{\text{power}(n, 2) = \text{power}(2 * k, 2)} [\text{substitution}]}{\text{power}(2 * k, 2) = 4 * \text{power}(k, 2)} [\text{algebra}]}{4 * \text{power}(k, 2) = 2 * (2 * \text{power}(k, 2))} [\text{factoring}]}{\exists m \bullet \text{power}(n, 2) = 2 * m} [\exists \text{intro with } m = 2 * \text{power}(k, 2)]}{\frac{\text{even}(\text{power}(n, 2))}{\text{even}(n) \Rightarrow \text{even}(\text{power}(n, 2))}} [\text{definition of even}]$$

Main theorem: If  $n^2$  is odd, then  $n$  is odd

$$\begin{array}{c}
\frac{}{\text{even}(\text{power}(n, 2))} \text{ [Lemma1]} \\
\frac{}{\text{odd}(\text{power}(n, 2)) \wedge \text{even}(\text{power}(n, 2))} \text{ [contradiction]} \\
\frac{\text{false}}{\text{odd}(n)} \text{ [false-elim]} \quad \frac{}{\text{odd}(n)} \text{ [identity]} \\
\frac{}{\text{odd}(n)} \text{ [\vee -elim]} \\
\frac{}{\text{odd}(\text{power}(n, 2)) \Rightarrow \text{odd}(n)} \text{ [\Rightarrow -intro}^{[1]}]
\end{array}$$

Proof by contrapositive using lemma.

### Example 9 : Proof by Minimal Counterexample

Prove: All natural numbers  $n \geq 1$  satisfy  $P(n)$

$$\begin{array}{c}
\frac{}{\mathbb{P} 1} \text{ [base case proved separately]} \quad \frac{}{\forall k \bullet k \geq 1 \wedge k < m \Rightarrow \mathbb{P} k} \text{ [minimality of m]} \\
\frac{}{\neg (\mathbb{P} m) \wedge \mathbb{P} 1} \text{ [contradiction]} \quad \frac{\mathbb{P}(m-1)}{\mathbb{P} m} \text{ [by inductive step from } P(m-1)] \\
\frac{}{\text{false}} \text{ [contradiction]} \quad \frac{}{\neg (\mathbb{P} m) \wedge \mathbb{P} m} \text{ [contradiction]} \\
\frac{}{\text{false}} \text{ [contradiction]} \quad \frac{}{\text{false}} \text{ [contradiction]} \\
\frac{}{\text{false}} \text{ [\vee -elim]} \\
\frac{}{\forall n \bullet n \geq 1 \Rightarrow \mathbb{P} n} \text{ [\neg -intro}^{[2]}] \\
\frac{}{\text{true} \Rightarrow (\forall n \bullet n \geq 1 \Rightarrow \mathbb{P} n)} \text{ [\Rightarrow -intro}^{[1]}]
\end{array}$$

Minimal counterexample combines well-ordering with contradiction.

### Example 10 : Proof by Invariant

Prove: A loop maintains invariant  $I$

Initialization:

$$\frac{\frac{}{\ulcorner \text{initial\_state} \urcorner^{[1]}}}{\text{invariant}(\text{initial\_state})} \text{ [verification]} \\
\frac{}{\text{initial\_state} \Rightarrow \text{invariant}(\text{initial\_state})} \text{ [\Rightarrow -intro}^{[1]}]$$

Preservation:

$$\frac{\frac{\frac{}{\ulcorner \text{invariant}(\text{before\_state}) \wedge \text{executes\_loop\_body} \urcorner^{[1]}}}{\text{invariant}(\text{before\_state})} \text{ [\wedge -elim-1]} \quad \frac{}{\text{executes\_loop\_body}} \text{ [verification]}}{\text{invariant}(\text{after\_state})} \text{ [\wedge -elim-2]} \\
\frac{}{(\text{invariant}(\text{before\_state}) \wedge \text{executes\_loop\_body}) \Rightarrow \text{invariant}(\text{after\_state})} \text{ [\Rightarrow -intro}^{[1]}]$$

Termination:

$$\begin{array}{c}
\frac{\frac{\frac{\lceil \text{loop\_terminates} \rceil^{[1]}}{\text{invariant}(\text{termination\_state})} \text{ [by preservation]}}{\text{invariant}(\text{termination\_state}) \wedge \text{termination\_condition}} \text{ [\wedge -intro]}}{\frac{\text{desired\_property}}{\text{loop\_terminates} \Rightarrow \text{desired\_property}} \text{ [\Rightarrow -intro}^{[1]}]} \text{ [logic]}
\end{array}$$

### Example 11 : Proof by Diagonalization

Prove: The set of real numbers is uncountable

$$\begin{array}{c}
\frac{\frac{\frac{\frac{\lceil \text{countable}(\text{reals}) \rceil^{[2]}}{\exists f \bullet \text{enumeration}(f, \text{reals})} \text{ [definition of countable]}}{\text{diagonal\_construction}(r)} \text{ [diagonal method]}}{\frac{\forall n \bullet r \neq \text{apply}(f, n)}{\text{not\_in\_range}(r, f)} \text{ [previous line]}} \text{ [by construction, differs at nth digit]}}{\frac{\text{not\_in\_range}(r, f) \wedge \text{enumeration}(f, \text{reals})}{\text{false}} \text{ [contradiction]}} \text{ [contradiction]} \\
\frac{\text{uncountable}(\text{reals})}{\text{true} \Rightarrow \text{uncountable}(\text{reals})} \text{ [\neg -intro}^{[2]}] \text{ [\Rightarrow -intro}^{[1]}]
\end{array}$$

Cantor's diagonal argument (outline).

### Example 12 : Constructive Proof Pattern

To constructively prove:  $\exists x \bullet \mathbb{P} x$

Strategy: 1. Explicitly construct a witness  $w$  2. Verify  $\mathbb{P}(w)$  holds 3. Conclude  $\exists x \bullet \mathbb{P} x$  with  $x = w$

Example: Prove  $\exists n : \mathbb{N} \bullet n > 100 \wedge n$  is even

$$\begin{array}{c}
\frac{\frac{\frac{\lceil \text{true} \rceil^{[1]}}{\text{witness\_construction}(102)} \text{ [construction]}}{\frac{102 > 100}{102 = 2 * 51} \text{ [arithmetic]}} \text{ [arithmetic]}}{\frac{\text{even}(102)}{102 > 100 \wedge \text{even}(102)} \text{ [definition]}} \text{ [\wedge -intro]} \\
\frac{\text{even}(102)}{102 > 100 \wedge \text{even}(102)} \text{ [\wedge -intro]} \\
\frac{\text{even}(102)}{\exists n \bullet n > 100 \wedge \text{even}(n)} \text{ [\exists intro with n = 102]} \\
\frac{\exists n \bullet n > 100 \wedge \text{even}(n)}{\text{true} \Rightarrow (\exists n \bullet n > 100 \wedge \text{even}(n))} \text{ [\Rightarrow -intro}^{[1]}]
\end{array}$$

### Example 13 : Proof Composition

Combine multiple proof techniques:

Theorem: Property  $\mathbb{P}$  holds for all cases

Overall strategy: Case analysis + Induction + Contradiction

